

Logic Design Demo Board

Description

This is a logic gate demo board design for high school students. The demo board is implement with six MOSFETs. The demo board has 3 NMOS and 3 PMOS. They can be combined into 5 different kinds of logic gates.. Also, an Arduino can be connected to the demo board as inputs, and the LED can visualize the output logic state.

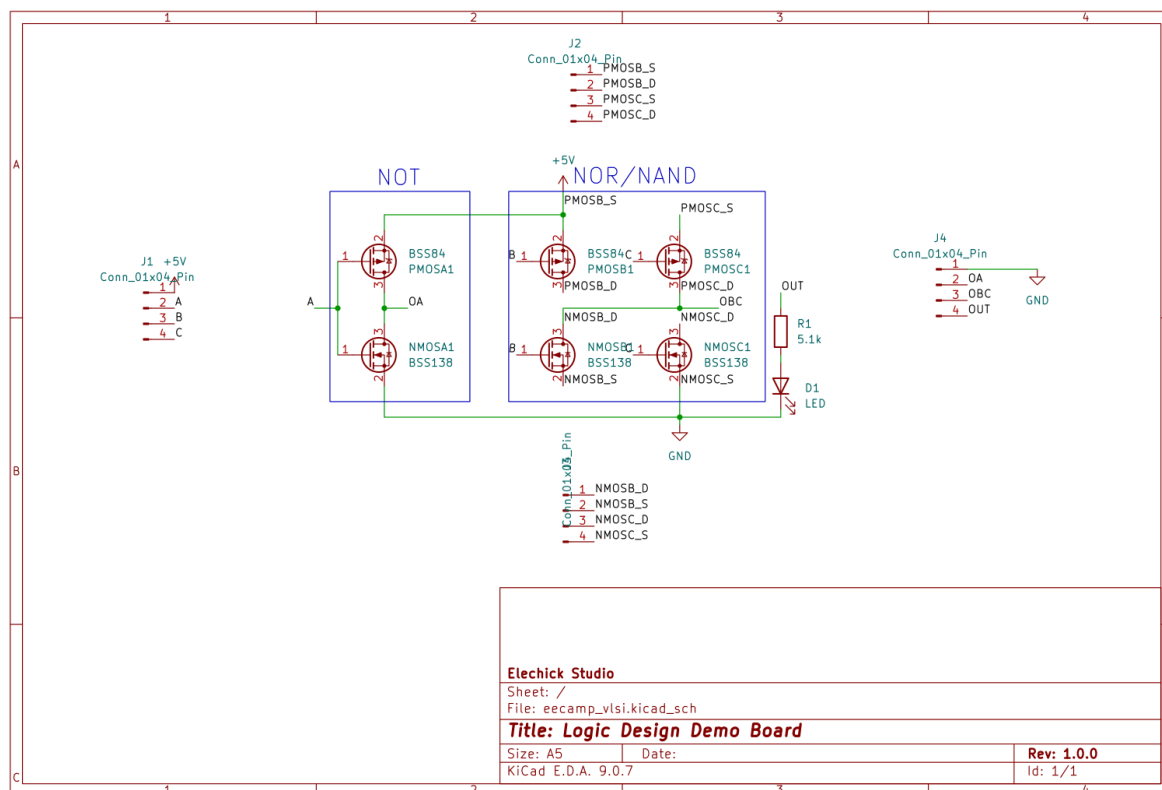


Figure1: Schematic

Pin Configuration

Pin name	Description
5V	Voltage supply
A	Data input for Inverter (PMOSA and NMOSA)
B	Data input 1 for the 4-MOSFET block (PMOSB/C, NMOSB/C).
C	Data input 2 for the 4-MOSFET block (PMOSB/C, NMOSB/C).
GND	Ground
OA, OBC	Raw logic output from the MOSFET circuit.
OUT	Output to LED.
PMOSx-y, NMOSx-y	“x” is the group letter (A/B/C), “y” is the terminal name (G for Gate, S for Source, D for Drain).

Reference Logic Gates Wiring

1) NOT Gate

- a) Uses Group A MOSFETs to invert input A.
- b) Connections:
 - 1) Connect OA to OUT.
 - 2) Apply input signal to A.

2) NAND Gate

- a) PMOS in parallel to 5V, NMOS in series to GND using Groups B and C.
- b) Connections:
 - 1) Connect PMOSB_S to PMOSC_S.
 - 2) Connect NMOSB_D to PMOS_D.
 - 3) Connect NMOSB_S to NMOSC_D.
 - 4) Connect OBC to OUT.
 - 5) Apply input signal to B and C.

3) NOR Gate

- a) PMOS in series to 5V, NMOS in parallel to GND using Groups B and C.
- b) Connections:
 - 1) Connect PMOSB_D to PMOSC_S.
 - 2) Connect NMOSC_D to NMOSB_D.
 - 3) Connect NMOSB_S to NMOSC_S.
 - 4) Connect OBC to OUT.
 - 5) Apply input signal to B and C.

4) AND Gate

- a) Connect the output of the NAND Gate into the input of the NOT Gate.
- b) Connections:
 - 1) Configure Groups B and C as NAND Gate.
 - 2) Instead of routing OBC directly to the LED(OUT), connect the NAND output OBC to input A.
 - 3) Connect OA to OUT.

5) OR Gate

- a) Connect the output of the NOR Gate into the input of the NOT Gate.
- b) Connections:

- 1) Configure Groups B and C as NOR Gate.
- 2) Instead of routing OBC directly to the LED(OUT), connect the NOR output OBC to input A.
- 3) Connect OA to OUT.

Example Code

```
// Define MCU pins connected to the Demo Board Pin headers
const int inputPin1 = 2; // Connects to B (or A for NOT gate testing)
const int inputPin2 = 3; // Connects to C

void setup() {
    pinMode(inputPin1, OUTPUT);
    pinMode(inputPin2, OUTPUT);
}

void loop() {
    // Test Case 1: 0 0
    digitalWrite(inputPin1, a);
    digitalWrite(inputPin2, b);
    delay(1000);

    // Test Case 2: 0 1
    digitalWrite(inputPin1, a);
    digitalWrite(inputPin2, b);
    delay(1000);

    // Test Case 3: 1 0
    digitalWrite(inputPin1, a);
    digitalWrite(inputPin2, b);
    delay(1000);

    // Test Case 4: 1 1
    digitalWrite(inputPin1, a);
    digitalWrite(inputPin2, b);
    delay(1000);
}
```

Revision History

PCB Revision	Datasheet Revision	Description
1.0	1.0.0	Initial version.